ARAB YOUTH SPACE HACKATHON 2026 · THE 813 CHALLENGE

Palm2Soil

Satellite-guided salinity remediation using date palm waste

409
field parcels ranked

426
spectral bands, 376–2499 nm

1
notebook, start to finish

THEME
Theme 2 — Precision Agriculture & Crop Intelligence

COUNTRY
United Arab Emirates

TEAM
Team Palm2Soil

DATE
October 2026

A ranked inspection-priority list for farm parcels, built end to end from one open hyperspectral acquisition.

Demonstrated on Planet Tanager-1 · 426 bands · CC-BY-4.0 · all figures are real pipeline outputs

THE PROBLEM AND WHO IS AFFECTED

Salinity removes farmland before anyone can see it

Decades of groundwater irrigation have left large areas of UAE farmland with soil salinity that cuts yield long before any damage is visible. Salinity does not announce itself: by the time a farmer can see it, the soil has been degrading for years.

- Farmers lose yield to a cause they cannot identify, and keep irrigating with groundwater that is already saline — so salt keeps accumulating.
- The authority holding the remediation budget has thousands of farms and no current evidence base for deciding which to treat first. Allocation is effectively guesswork.
- Separately, date palm pruning waste is treated as a disposal cost, although UAE trials show biochar made from it improves water retention and microbial health in sandy, salt-affected soil.

THE LAST FULL SURVEY WAS 2009

~90% of farms around Al Ain were saline

43% above 9.4 dS/m

26% in the 6.25–9.4 dS/m band, where soil amendment is most likely to help

Environment Agency – Abu Dhabi assessment, ~50,000 soil samples (Abdelfattah et al., 2009). It has not been repeated at that scale in seventeen years, because ground sampling at that density is not economically repeatable. Satellite measurement is.

The gap we address Budget exists. An evidence base for where to spend it does not — and ground sampling at survey density is not economically repeatable.

THE BUSINESS USE CASE AND THE END USER

Where to inspect first, and what each tier receives

THE CUSTOMER

The authority allocating agricultural remediation budget across thousands of farms.

They buy a defensible order of inspection — a ranked list with data-quality figures attached to every row, so a decision can be audited. The farmer is the second user: the field app tells them which of their blocks was flagged and records what they find.

DETECT

One hyperspectral acquisition, masked for cloud and no-data, segmented into field parcels.

BUILT IN THIS PoC

>

PRIORITISE

Each parcel ranked by canopy–chlorophyll divergence, with a confidence flag.

BUILT IN THIS PoC

>

INTERVENE

Each tier matched to an action: biochar and organic amendment where stress is moderate, drainage or leaching where severe, crop substitution where recovery is not viable.

INCUBATION

>

VERIFY

Later scans of the same blocks measure whether treated farms recover against untreated controls — and feed that back into the ranking.

INCUBATION

The defensible product is the targeting layer. Biochar is what one tier receives — its performance depends on pyrolysis conditions and application rate, and some biochar carries elevated conductivity of its own, so it is treated as one matched intervention, not a universal fix.

DATA

One open hyperspectral acquisition, and what it is compared against

	SOURCE
Imagery	Planet Tanager-1, orthorectified surface reflectance
Scene	20250608_091605_90_4001 — Çumra, Konya, Türkiye, 8 June 2025
Sensor	426 bands, 376–2499 nm, 30 m, HDF5
Access	Planet open STAC catalogue · licence CC-BY-4.0
Scene quality	2% cloud · 68.7% valid pixels after masking
Seasonal context	Sentinel-2 L2A via Planetary Computer, 25 dates
S-2 responses	ESA COPE-GSEG-EOPG-TN-15-0007 S2-SRF v5, measured

Is hyperspectral actually necessary here?

Yes, and we tested it rather than asserting it. Sentinel-2 was simulated by integrating the Tanager spectrum through ESA's measured response functions — same ground, same instant, same atmosphere, so spectral sampling is the only variable.

1500–2500 nm	TANAGER-1	SENTINEL-2
Bands in range	**201**	2
1931–2123 nm salt window	**39**	6.2% of response
1667–2092 nm	**86**	zero — blind gap
2316–2450 nm	**27**	zero — B12 ends

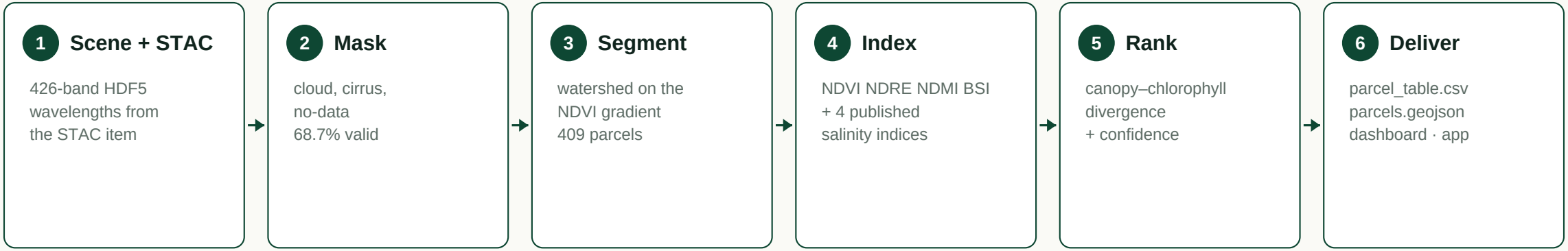
Band counts are sensor design — true regardless of scene.

The physics, stated correctly

Pure halite has NO diagnostic absorption in the VNIR–SWIR. We never claim direct salt detection. What is detectable is indirect: hydration features in salt crusts and associated evaporites such as gypsum, near 1450, 1850, 1950, 2200 and 2340 nm — absorptions tens of nanometres wide, which a 173 nm-wide band cannot resolve.

APPROACH

Scene to ranked list, in one notebook that runs end to end



The return leg — built, and in the package

The field app exports its ground checks as JSON. import_observations.py validates each record against the parcel table and writes one row per observation with that parcel's satellite measurements attached. Records that cannot be matched — unknown parcel, wrong scene, wrong run, malformed timestamp — are reported individually and the command exits non-zero. This joins; it does not calibrate.

Runs offline
the bundled 128 × 128 px sample needs no network

~2 minutes
end to end on the sample, Python 3.12

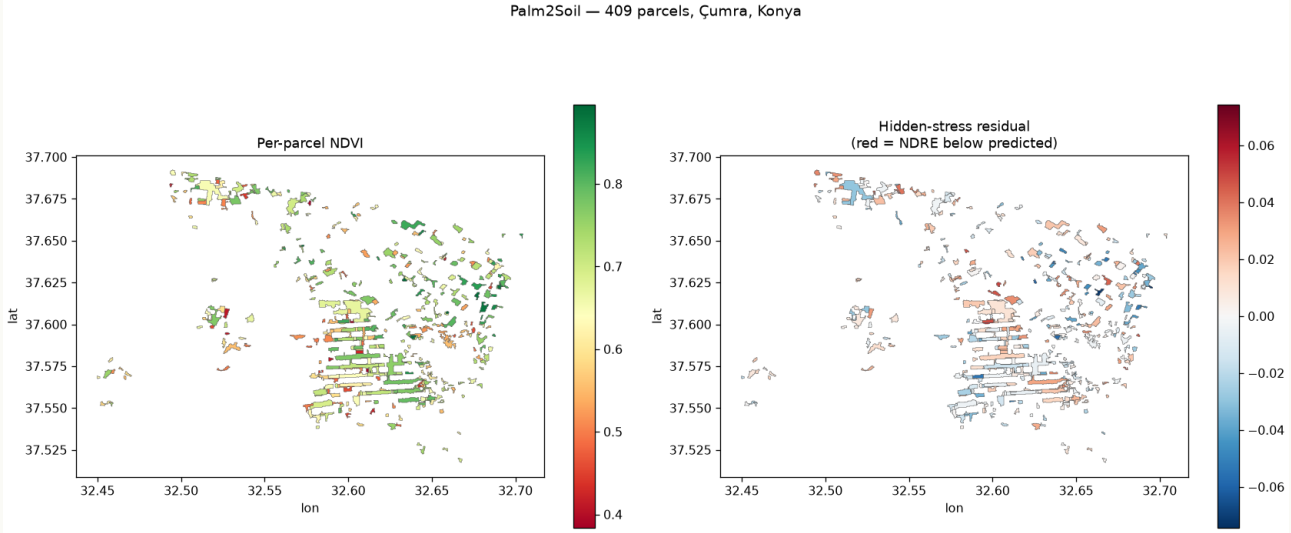
Fingerprinted
every row carries a run_id; parcel ids are only unique within a run

Regression-tested
33 smoke checks and 31 unit tests ship with it

EXAMPLE OUTPUT

409 field parcels, each with its own measurements and ranking

Left: greenness. Right: the ranking quantity — how far each parcel's red-edge chlorophyll signal (NDRE) falls below the level its own greenness (NDVI) predicts. Red is more divergent. The two maps do not look alike, which is the point: the ranking is not re-describing greenness.



parcel_check.png — drawn by the notebook. Scene extent 22.5 × 19.5 km, 30 m pixels, Çumra, Konya, 8 June 2025.

409
PARCELS MAPPED
2,986 ha · median 4.0 ha

345
USABLE FOR RANKING
excludes low-confidence and merged

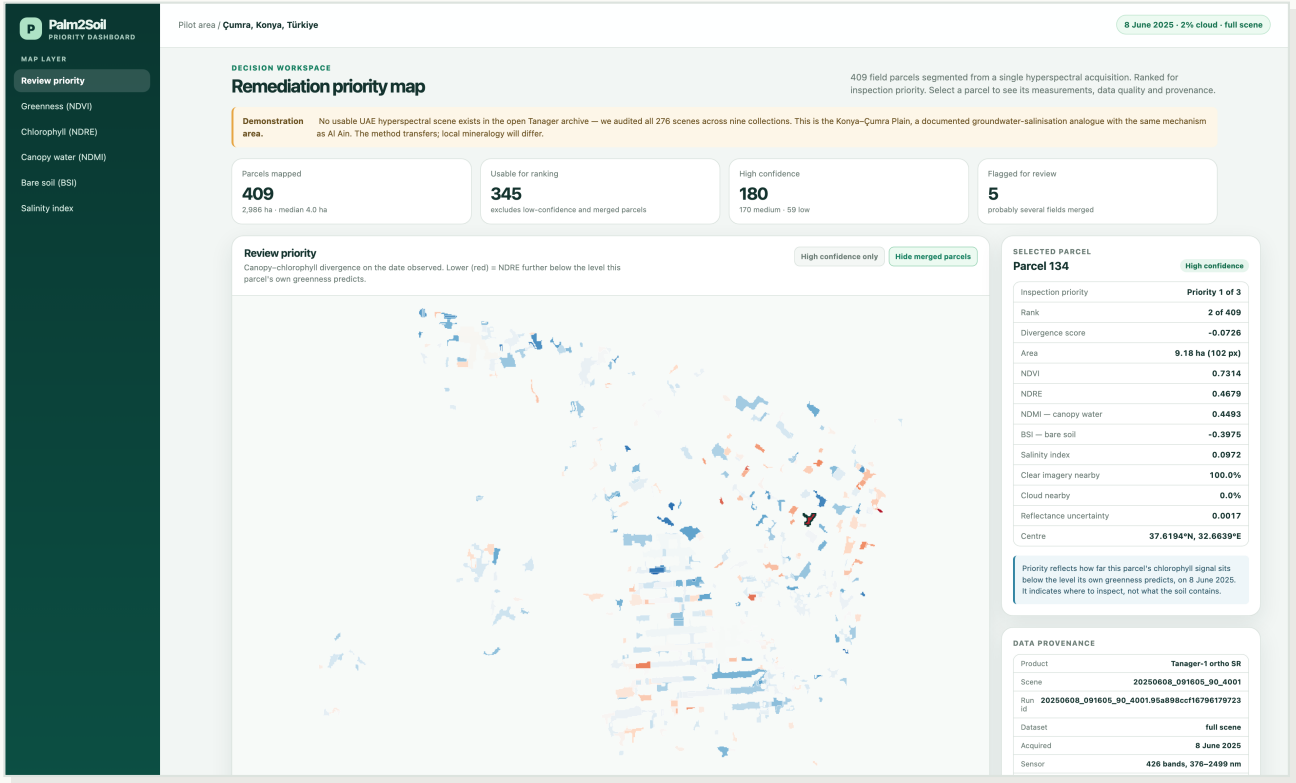
180
HIGH CONFIDENCE
170 medium · 59 low

5
FLAGGED FOR REVIEW
probably several fields merged

EXAMPLE OUTPUT

The authority dashboard — the actual deliverable

A self-contained HTML file, no server and no internet. Select any parcel to see its measurements, its data quality and the provenance of the run that produced it.



Palm2Soil_Dashboard.html, rendered from the submitted file.

What a judge can check

- Five map layers over the same parcels — priority, NDVI, NDRE, NDMI, BSI and a salinity index.
- Every selected parcel shows clear imagery nearby, cloud nearby and the provider's reflectance uncertainty.
- Low-confidence and merged parcels are excluded from priority and rank, not quietly scored anyway.
- The run id is printed on screen, so a figure can always be traced to the run that made it.

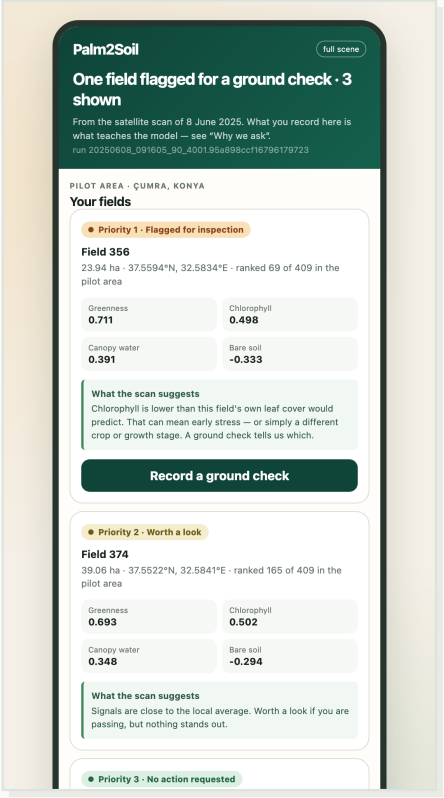
STATED ON THE DASHBOARD ITSELF

Priority reflects how far a parcel's chlorophyll signal sits below the level its own greenness predicts, on 8 June 2025. It indicates where to inspect, not what the soil contains.

EXAMPLE OUTPUT

The field app and the return leg

The second half of the loop: the app puts the flagged fields in front of whoever walks them, and the importer brings what they find back to the measurements.



Palm2Soil_FieldApp.html

Field app

Shows one field per priority group with plain-language wording — "chlorophyll is lower than this field's own leaf cover would predict" — and never asserts salinity. Records are stored against the run id, so a record made against one run cannot be read back into another. If the browser refuses to store, the app says so rather than losing the record silently.

import_observations.py

Joins exported records to the parcel table and writes observations_joined.csv. It rejects, individually and loudly, a boolean or fractional field id, a non-finite id, an unknown parcel, a record from another scene or another run, and a malformed timestamp. Unmatched records go to their own file and the exit status is non-zero.

WHY THIS IS STRICT

An unmatched record almost always means a real mistake — a stale export, or a table regenerated since the observation was made. Silence would hide it, and this joined table is exactly the input a dS/m calibration would start from.

Try it against the two bundled example files

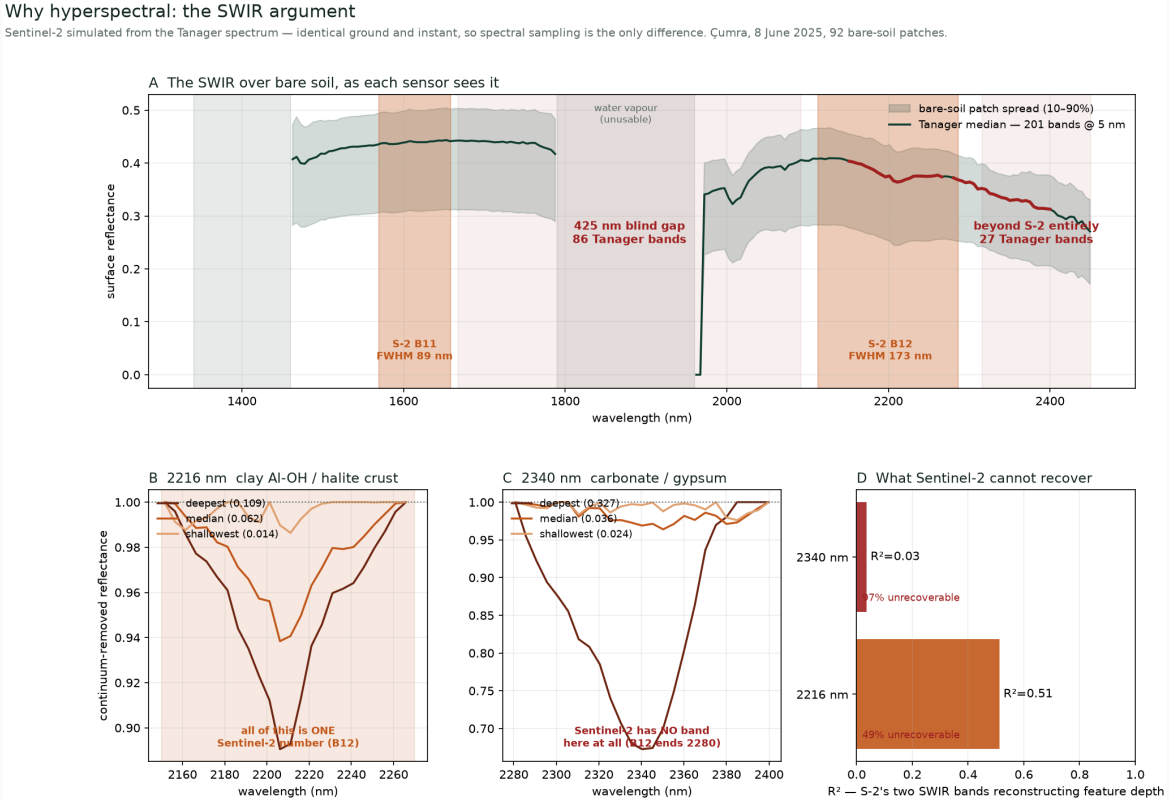
```
python import_observations.py examples/input/sample_field_records.json \
       --data-dir examples/output
```

sample_field_records_with_errors.json exercises every rejection path, and the command exits non-zero.

EXAMPLE OUTPUT

Why hyperspectral — measured, not asserted

Sentinel-2 simulated from the Tanager spectrum through ESA's measured response functions: identical ground, identical instant, so spectral sampling is the only difference. 92 bare-soil patches.



A — the two sensors over the same soil

Tanager samples 1500–2500 nm with 201 contiguous bands. Sentinel-2 has two, with a 425 nm gap between them and no response at all beyond 2316 nm.

B and C — the features themselves

The 2216 nm clay/halite-crust feature is about 40 nm wide; all of it collapses into one Sentinel-2 number. The 2340 nm carbonate/gypsum feature sits outside the instrument entirely.

D — what cannot be recovered

Regressing feature depth on the simulated Sentinel-2 bands: R^2 0.51 at 2216 nm, R^2 0.03 at 2340 nm. Roughly half and nearly all of the variance is unrecoverable.

Detection and crop monitoring need different pixels. Salt crusts are a soil-surface phenomenon, so these features are measured on bare ground; over a closed canopy the SWIR carries leaf water, not soil chemistry. That falls out of the physics and the product design reflects it.

swir_absorption_argument.png — reproducible from src/make_swir_figure.py on the full scene.

VALIDATION

What we checked, and what the checks showed

Two checks that the ranking measures something real

- Spatially coherent. Neighbouring parcels have correlated scores (r = +0.23). Noise would be spatially random.
- Physically consistent. The most-divergent decile is drier (NDMI –0.045), barer (BSI +0.036) and more saline-looking (+0.016) than the least divergent — while being only marginally less green (NDVI –0.018). Standard greenness separates the two groups least of all, which is the whole case for not simply ranking on NDVI.

Reproducibility is part of the claim

- 33 end-to-end smoke checks: the notebook executes, ids agree between table and GeoJSON, every geometry is topologically valid, every coordinate falls inside the sample footprint, and both apps rebuild from the outputs.
- 31 unit tests over the pipeline, the apps and run identity.
- Every published number traces to a script in src/.

Two results that went against us — both kept

The red-edge justification did not hold.

Our application argued hyperspectral on red-edge position shifts. Sentinel-2 reconstructs red-edge position at R^2 0.68, and its NDRE separated an independent grouping better than either red-edge metric. We rebuilt the case on SWIR absorption, where the evidence is far stronger.

The score does not predict the season.

Tested across 25 Sentinel-2 dates over the same parcels: |r| is 0.25 or below, with signs opposite to a stress interpretation. On one acquisition, divergence cannot be cleanly separated from crop type and growth stage. So the score is an inspection priority, not a forecast.

Both are reproducible from the notebook in minutes, and each one led to a better-specified design.

WHAT THIS PROOF OF CONCEPT ESTABLISHES

- A spatially coherent canopy–chlorophyll divergence signal is measurable at field-parcel level.
- Soil absorption features exist that broadband sensors demonstrably cannot resolve.
- On the date measured it tracks independent moisture and soil-exposure indicators, while being nearly invisible to greenness.
- The same parcels track through a full season from free imagery, and the pipeline reproduces end to end from open data.

LIMITATIONS

What this proof of concept does not establish

Stated plainly, because a proof of concept that overstates its reach is worth less than one that knows its boundaries.

No absolute salinity

All ranking is relative within a scene. Converting to dS/m needs paired ground measurements at known locations; the 2009 EAD assessment reaches us only as regional percentages, and a per-pixel map cannot be validated against a summary statistic. Intervention tiering is a designed mechanism, not a calibrated one.

No seasonal prediction

Measured, not assumed — see the previous slide. The crop-stage confound is the most likely cause and needs multi-date acquisition or ground truth to resolve.

Parcels are not cadastral

They are segmented from the imagery by watershed on the NDVI gradient and approximate management units. 5 of 409 are flagged as probably several merged fields and excluded from the working set.

Single acquisition

Tanager imaged this footprint once in the entire open archive — a real operational constraint of the sensor. Seasonal tracking is therefore demonstrated with Sentinel-2, which is also the honest architecture: a cheap broadband monitor, with hyperspectral where the narrow features are.

No treatment effect

No treatment occurred in this scene, so recovery of treated against untreated blocks is designed rather than shown.

Salinity indices are not calibrated

The four published indices are verified as formulas and computed on broadband-equivalent reflectance, but none is tied to a measured dS/m value. They are descriptive context; the ranking does not depend on them.

Why the demonstration is in Türkiye, not the UAE

There is no usable UAE hyperspectral scene in the open Tanager archive. We established that by auditing all 276 scenes across all nine collections rather than assuming it: three fall in MENA, all above 50% cloud, and the single UAE scene is 51% cloud and 25.8° off-nadir over Al Dhafrah. We therefore demonstrate on the Konya–Çumra Plain — a semi-arid endorheic basin where about 90% of irrigation water is groundwater and the water table is falling roughly 1 m a year. The sampling argument transfers unchanged because it concerns instrument design; the specific mineral features will differ with local soil chemistry.

IMPACT

An evidence base for spending that currently has none

Food security is a stated national priority, yet the productive capacity of UAE farmland is being eroded by a process that is invisible until yield has already fallen. That is exactly the kind of slow, distributed problem ground surveys are too expensive to catch and satellites are well suited to.

We chose it because it is unglamorous and consequential. Land degradation does not generate headlines the way flooding or air quality do, but it permanently removes agricultural capacity from a country with very little of it.

Closing the loop with date palm waste turns a disposal cost into a remediation input — the circular-economy logic the UAE has committed to, and rarely gets to demonstrate this concretely.

Allocation

Remediation spending gets a defensible order of inspection, auditable row by row, in place of guesswork.

Circularity

A route for diverting date palm pruning waste from disposal into soil improvement, matched to a specific severity tier.

Transferability

The method moves directly to Oman, Saudi Arabia, Bahrain and Kuwait — the same groundwater salinisation on comparable soils.

CONTRIBUTES TO

SDG 2 · 6 · 12 · 13 · 15

Zero hunger · clean water · responsible consumption · climate action · life on land.

NEXT STEPS

What incubation is for

In priority order. The first item unblocks most of the rest — it is the single thing standing between a relative ranking and a calibrated measurement.

1. **Georeferenced ground truth**
   Paired soil samples at known locations in the pilot area, enough to fit and test a dS/m model. Aggregate regional percentages cannot validate a per-pixel map.
   Unblocks absolute salinity

2. **Hyperspectral coverage of Al Ain**
   EnMAP tasking is being pursued. A UAE scene replaces the analogue and lets the mineralogy argument be made on local soil chemistry.
   Removes the analogue caveat

3. **Multi-date acquisition**
   Resolves the crop-stage confound that stopped the single-date score predicting the season, and turns inspection priority into something closer to a forecast.
   Addresses the measured negative

4. **Treatment trial with controls**
   Treated and untreated blocks, tracked across a season, to measure recovery and close the verification leg that is currently designed rather than shown.
   Proves the loop

5. **Cadastral boundaries**
   Replace watershed-segmented parcels with registry boundaries so a row corresponds to a farm an authority can actually write a cheque against.
   Operational readiness

IN THE SUBMISSION PACKAGE

Everything here is runnable

07_palm2soil_salinity_prioritisation.ipynb
The pipeline, end to end. Run All, about two minutes, no network.

requirements.txt
Pinned, with an Intel-macOS note for the rasterio and pyproj wheels.

examples/input/
A 128 × 128 px crop of the scene that runs offline, the STAC item, and field records including deliberately broken ones.

examples/output/
Full-scene parcel table, GeoJSON, figures and the statistics behind every number in this deck.

Palm2Soil_Dashboard.html
The authority dashboard, self-contained.

Palm2Soil_FieldApp.html
The field capture app, self-contained.

smoke_test.py · tests/
33 end-to-end checks and 31 unit tests.

src/
Every script that produced every published figure.

TEAM Syeda Amsha Nasar · Zahny Fatima · Huda Ibrahim · Hanan Ibrahim · Marwa Khot

Palm2Soil ranks parcels relative to others in the same scene. It does not measure salinity in dS/m. We would rather say so here than be asked.

Imagery: Planet Tanager-1, CC-BY-4.0 · Sentinel-2: ESA/Copernicus via Microsoft Planetary Computer · Code: MIT